

Oil Spill Detection & Vessel-Source Attribution using SAR + AIS

SIH 2026 – Problem Statement 26143 (NTRO) – Software / Disaster
Management – Deep-Dive Research Dossier

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This document follows the exact 5-step framework you gave — Understand → Research → Gap → RCA → Score — and then adds a full technical/mathematical annexure (Section 6) covering the physics, algorithms, and equations that a real solution to this PS would need, from basic to advanced. Every claim about an existing tool, dataset, paper, or startup below is grounded in current sources (Aug 2026); the equations in Section 6 are standard results from the oil-spill-science and SAR-remote-sensing literature, with the caveats the literature itself attaches to them (empirical constants vary by source, models are known to under/over-predict in specific regimes, etc.) — I’ve kept those caveats in rather than presenting textbook formulas as exact truths, because that nuance is exactly what separates a strong PS submission from a shallow one.

1. Understand the Problem

1.1 What the PS is actually asking for

Two coupled tasks, not one:

1. **Detection** — find oil on the sea surface from satellite imagery (mainly SAR, optionally optical/EO), and calculate its geometric properties (area, shape) and, if feasible, its age.
2. **Attribution** — reconstruct where and when the oil was discharged (forward/backward drift + weathering), then cross-reference that against historic AIS vessel-traffic data to rank which vessel(s) most likely caused it, filtering out irrelevant traffic and scoring suspects on proximity, trajectory, and behavioral anomalies.

Plus a usable visual interface. Both halves are individually well-studied; doing them *together*, well, and *tuned to Indian waters* is the actual ask.

1.2 Who actually faces this problem — stakeholder map

Primary (own the pain directly):

- **Indian Coast Guard (ICG)** — the statutory central coordinating authority for marine pollution response under India's National Oil Spill Disaster Contingency Plan (NOS-DCP). They need fast detection + a defensible list of suspects to act on.
- **NTRO** — the sponsoring org itself; maritime domain awareness (MDA) and strategic surveillance of India's coastline/EEZ is core to its mandate, of which oil-spill attribution is one instance of a broader "who is doing what at sea" problem.
- **Directorate General of Shipping / Ministry of Ports, Shipping & Waterways** — enforces MARPOL Annex I (the international regime preventing pollution from ships) in Indian waters and conducts port-state-control action against violators.
- **Ministry of Environment, Forest & Climate Change / State Pollution Control Boards** — assess environmental damage and pursue liability.

Secondary (affected, not directly responsible for response):

- Coastal fishing communities (livelihoods and health)
- Marine-protected-area managers (e.g., Gulf of Mannar, Gulf of Kutch mangroves/reefs)
- Shipping companies, P&I clubs, and insurers — note this cuts *both ways*: a good attribution system also protects innocent vessels from being wrongly implicated
- National Green Tribunal / judiciary — any output that's meant to support legal action needs to be evidentially sound, not just "the ML model said so"

Tertiary: coastal tourism economy, the general public, India's international treaty obligations (MARPOL, OPRC 1990).

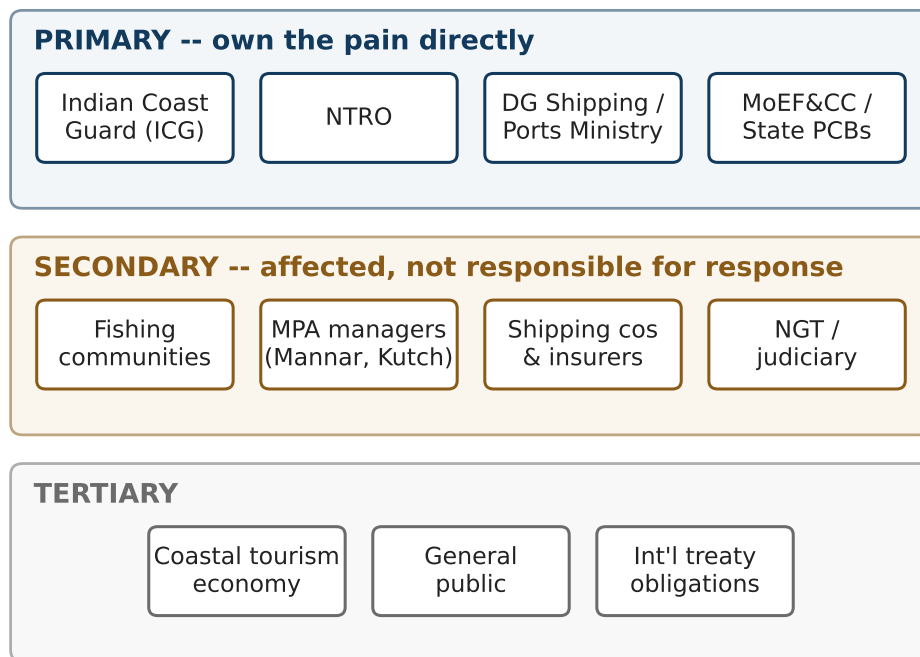


Figure 1: Stakeholder map for PS 26143 – who actually faces this problem.

1.3 The real pain point (not just “oil spills are bad”)

There are two distinct, compounding pain points:

- **Detection latency & coverage:** India’s EEZ is roughly 2.4 million km² with a ~7,500 km coastline. Continuous manual surveillance is impossible. Small, chronic “bilge dumping” spills — which happen far more often than headline tanker disasters — are specifically timed by polluters to exploit exactly this gap (open sea, night, bad weather).
- **Attribution difficulty:** even when a slick is detected, oil keeps drifting, spreading, and weathering between discharge and detection. By the time anyone looks, several vessels may have transited the area — a genuine many-to-one / one-to-many matching problem. Without a systematic, physics-grounded reconstruction, culprits can’t be pinned down beyond a “well, this ship was nearby” hand-wave — which isn’t good enough to trigger inspection, let alone stand up legally.

1.4 Why the problem exists — four structural reasons

1. **Physical ambiguity:** a genuine oil film and several completely natural phenomena (biogenic slicks, low-wind zones, current-shear fronts, rain cells) look nearly identical in SAR — both just dampen the sea surface. This is a root-cause item, not a solvable-with-more-effort item (see RCA, §4).

2. **Data fragmentation:** satellite tasking (ISRO), AIS (DG Shipping / ICG), and met-ocean forcing data (IMD, INCOIS) live in separate systems with no single automated Indian pipeline stitching them together the way EMSA's CleanSeaNet+SafeSeaNet does for Europe.
3. **AIS is voluntary and spoofable for exactly the vessels most likely to pollute illegally:** SOLAS mandates AIS broadly for vessels over 300 GT on international voyages, but a large share of Indian coastal, fishing, and small non-SOLAS traffic either doesn't carry continuously-broadcasting AIS or can switch it off — a genuine structural (not engineering) gap.
4. **Institutional capacity:** the existing workflow (an analyst manually reading SAR dark spots, then manually pulling AIS logs) doesn't scale to continuous, day/night, all-weather coverage of the whole coastline, and doesn't naturally produce a standardized, auditable confidence score.

1.5 What “success” would actually change

- Detection latency: from “unknown until it washes ashore or a patrol notices” → near-real-time flagging within one satellite revisit window.
 - Attribution confidence: from informal manual guesswork → a ranked, explainable shortlist of suspect vessels with quantified, evidence-backed confidence — usable to trigger boarding/inspection or legal notice.
 - Deterrence: EMSA's own decade-plus of CleanSeaNet operation shows a documented year-on-year decline in detected illegal discharges per unit area monitored — i.e., visible satellite-enforcement capability changes behavior on its own, before any single case is even prosecuted.
 - Cost/speed: less dependence on slow, expensive manual image-by-image review.
-

2. Research Existing Solutions

2.1 Government / intergovernmental platforms

- **EMSA CleanSeaNet (EU, operational since April 2007)** — the closest thing to a mature, production version of exactly this PS. Near-real-time SAR (plus some optical) oil-spill *and* vessel-detection service, delivering 3,000+ satellite images/year to EU member states, tightly integrated with the SafeSeaNet AIS network so that a detected vessel in the SAR image can be cross-checked against AIS tracks to identify a likely polluter. Documented outcome: a year-on-year *reduction* in detected spills per million km² monitored over the past decade — direct evidence that this kind of system deters illegal discharge, not just detects it after the fact.

- **NOAA Office of Response & Restoration (US)** — runs **GNOME** (General NOAA Operational Modeling Environment) + **ADIOS** (oil weathering model), the reference open-source Lagrangian particle-tracking trajectory-and-fate suite. Used for tactical spill response, strategic contingency planning, *and* post-spill source attribution (running the model backward from a known slick to the most likely origin). Fully open source (PyGNOME core, WebGNOME interface, on GitHub).
- **ISRO (India)** — Bhuvan GIS platform has integrated RISAT-1 SAR-derived oil-spill maps for Indian seas since at least 2014; more recently, **EOS-4** (India’s own C-band SAR satellite, launched Feb 2022) was used operationally to detect the oil spill from the MSC ELSA 3 container-ship capsized off the Kerala coast in May 2025 — i.e., India already has the raw sensing capability this PS needs; what’s missing is the automated attribution layer.
- **Indian Coast Guard / coastal surveillance** — a NavIC-based Vessel Tracking System already connects sub-20m fishing vessels (exactly the “dark,” non-AIS-mandated segment) into the Information Management and Analysis Centre (IMAC) for coastal security — a genuinely useful, underused data source for this PS’s Indian context.
- India’s statutory backbone: the **National Oil Spill Disaster Contingency Plan (NOS-DCP)**, with ICG as the central coordinating authority.

2.2 Deployed applications (the single most important comparison)

- **SkyTruth’s Cerulean** (public beta since ~2023, since integrated into Global Fishing Watch’s **Skylight** platform) is functionally *the same pipeline this PS is asking for, already built and running globally, open source*. It scans Sentinel-1 SAR imagery worldwide with a computer-vision model, outlines slick polygons with a confidence score, then cross-correlates each slick against AIS vessel tracks and a database of offshore infrastructure, ranking the top-3 probable sources per slick (“Source Profiles,” which also factor in AIS-off-event history). Code is public on GitHub; there’s a free map and an OGC-compliant API. Bellingcat’s open-source investigative guide to marine oil-spill detection cites Cerulean as a key public-interest tool.
- This is not a discouragement — it’s the single most useful benchmark to study before writing a line of code, because it tells you exactly where the *real* remaining gaps are (see §3).

2.3 Startups

- **PierSight Space (Indian)** — building a dedicated small-sat SAR + AIS constellation specifically for maritime domain awareness, oil-spill detection included. Won the US-India DIU/IDEX **INDUS-X Maritime Challenge**, whose technical

brief explicitly required fusing X-band radar, EO/low-light cameras, and satellite sensor streams to detect spills over 100,000 gallons in near-real-time. Has a demo mission planned on ISRO’s PSLV Orbital Experimental Module (POEM) and is actively partnering with the **Indian Coast Guard** for commercialization. If judges ask “how is this different from an existing Indian company already doing this,” PierSight is who they mean — know this cold.

- **Windward** (global “Maritime AI” company) — an enterprise vessel-risk-intelligence platform. Its Remote Sensing Intelligence module explicitly lists oil-spill detection as one use case, fusing SAR/EO/RF imagery with AIS and proprietary behavioral models to flag dark activity, AIS/GNSS spoofing, and ship-to-ship transfers; recently partnered with Vantor for persistent satellite tracking of “dark” (non-broadcasting) vessels. Closed/commercial, enterprise-priced, oil spill is a small feature of a much bigger platform — not a specialized, transparent tool.

2.4 Research literature

- **Krestenitis, Orfanidis, Ioannidis, Avgerinakis, Vrochidis & Kompatsiaris (2019)**, “Oil Spill Identification from Satellite Images Using Deep Neural Networks,” *Remote Sensing* 11(15):1762 — **the reference public benchmark** for this exact detection sub-task: Sentinel-1 SAR (VV polarization), 1,002 training + 110 test images (1250×650 px, 10 m resolution), 5-class semantic segmentation (sea surface, oil spill, look-alike, ship, land). Showed DeepLabv3+/U-Net outperforming earlier methods; 280+ citations; effectively the dataset every subsequent paper compares against.
- **Deep-SAR “SOS” dataset (Zhu et al., 2022)** — 21 raw SAR scenes (14 ALOS/PALSAR L-band HH-polarization + 7 Sentinel-1 C-band VV-polarization), cropped into 8,070 labeled 256×256 patches — a good complementary dataset for testing cross-sensor generalization.
- **Solberg et al.** established the canonical 3-step pipeline almost all subsequent work follows or reframes as one end-to-end model: (1) dark-spot detection, (2) feature extraction, (3) discrimination of oil vs. look-alike.
- **Attribution-specific literature:**
 - **Seatrack Web** (Ambjörn, 2008, Swedish Meteorological & Hydrological Institute) — operational Baltic-Sea tool combining backward-drift modeling with AIS mining; established the classic method of propagating a detected slick backward in time and finding whose AIS track intersects the backtracked position, but also flagged that backtracking uncertainty *grows* the longer the backtrack window is.
 - **Mouche et al.** proposed the *opposite* direction: forward-simulate a hypothetical continuous discharge along every candidate vessel’s actual route, and compare each simulated footprint against the observed slick — this

avoids compounding all uncertainty into one single backward run.

- A 2024 **bidirectional drift model** paper explicitly combines both directions specifically to resolve the many-to-one / one-to-many ambiguity that pure one-directional models suffer from in dense-traffic waters — this is presented as a still-open methodological question, not solved engineering.
- A dual-layer AIS+SAR framework paper uses **Isolation Forest** for unsupervised AIS behavioral-anomaly detection plus an **SVM** for satellite-image spill classification — built, in the authors’ own framing, because a mature SAR+AIS-based accountability system didn’t yet exist for their country. That framing maps almost exactly onto what NTRO is asking Indian teams to build.
- **Janeiro et al. (2017)** quantified how backtracking error grows with the length of the backtrack window, validated against real satellite-tracked drifter buoys — this is the kind of number you need if you want to honestly state “our attribution is reliable up to X hours after discharge, and degrades beyond that.”

2.5 Open-source tooling (huge time-savers for a 3-week sprint)

- **ESA SNAP (Sentinel Application Platform)** ships a built-in “Oil Spill Detection” operator: calibration → land-sea masking → adaptive-threshold dark-spot detection → cluster/size-based discrimination — a ready-made baseline or feature-extraction front end, free.
- **OpenDrift / OpenOil** (MET Norway, GPLv2, Python) — a generic Lagrangian trajectory framework; the OpenOil module implements 3D oil drift plus NOAA-derived weathering (evaporation, emulsification) physics, and can ingest Copernicus Marine Service current/wind forcing directly — the fastest realistic path to a working forward/backward drift-and-fate simulator without deriving every equation from scratch.
- **NOAA PyGNOME / WebGNOME / ADIOS** — the US government’s own open-source reference suite; the ADIOS Oil Database supplies real physicochemical properties (density, viscosity, distillation data) for hundreds of actual oil types, useful for realistic weathering parameters.
- **TorchGeo** (a PyTorch geospatial-ML library) has an open GitHub issue actively adding the Krestenitis dataset as a first-class loader — a sign this exact dataset/task is becoming a standard ML benchmark entry point, and that a decent baseline model is well within a hackathon’s reach.

2.6 Adjacent hackathon / student projects

“Oil spill detection using satellite + AIS with SVM/Random Forest” is a template idea that appears repeatedly across generic student ML-project listicles — meaning the

surface-level pitch is well-trodden and low-differentiation on its own; judges have likely seen shallow SVM/RF submissions before. I couldn't find that this exact NTRO-worded PS has run in a prior official SIH edition — it appears new to SIH in 2026 — but the underlying technical idea is not new to the research world; it maps directly onto 15+ years of EMSA/academic work. **This matters directly for the gap analysis:** “we combined satellite + AIS” cannot be your differentiator; it has to be the specific gaps below.

3. Find the Gap

Existing solution	Does well	Gap for <i>this</i> PS's Indian/NTRO context	Your improvement angle
EMSA CleanSeaNet	Operational, AIS-integrated, decade-long track record	EU-only; not tuned to Indian coastal traffic mix, monsoon sea-states, tropical biogenic look-alikes	Build/validate a look-alike discriminator on Indian-coast conditions (monsoon wind regimes, aquaculture/fishing-related films, river-plume fronts near major deltas)
NOAA GNOME/ADIOS, OpenDrift/OpenOil	Rigorous, open, reusable drift+weathering physics	Generic engines with no packaged “slick polygon + timestamp → ranked candidate vessels” workflow	Use OpenOil/GNOME-style physics as the <i>core engine</i> , then build an original, explainable multi-criteria vessel-ranking layer on top — this layer is the real novel-contribution surface

Existing solution	Does well	Gap for <i>this</i> PS's Indian/NTRO context	Your improvement angle
SkyTruth Cerulean	Near-identical end-to-end pipeline, open source, global, free, already has a “top-3 sources” ranking	Global/general-purpose; not India-EEZ-focused; doesn't leverage India's own EOS-4/RISAT/NavIC data; its ranking is a comparatively simple proximity/AIS-off heuristic — no published bidirectional-drift + behavioral-anomaly fusion	Localize to Indian SAR sources + Indian small-vessel/NavIC context; replace simple proximity ranking with a documented, weighted, explainable multi-factor score (drift-overlap + CPA/TCPA + AIS-gap + behavioral anomaly) with a full audit trail
PierSight	India-specific, hard-ware+software co-designed, ICG partnership already underway	It's a satellite-hardware company — a hackathon team cannot compete on sensors	Position your prototype as the <i>open, demonstrable analytics layer</i> that plugs into whatever SAR feed is available (ISRO or otherwise), not as a satellite operator

Existing solution	Does well	Gap for <i>this</i> PS's Indian/NTRO context	Your improvement angle
Windward	Mature dark-vessel/behavioral-anomaly detection, multi-sensor fusion, enterprise-grade	Proprietary, closed, expensive, general-purpose (oil spill is one small feature among many)	Make your scoring logic fully transparent/inspectable — valuable both for hackathon judging on “prototype feasibility” and for eventual government adoption, where explainability matters for legal defensibility
Krestenitis / SOS academic datasets & models	State-of-the-art accuracy on their own benchmark	Trained on European/Gulf-of-Mexico/Persian-Gulf scenes — may not generalize cleanly to Indian coastal SAR statistics and small-craft clutter without adaptation	Fine-tune/domain-adapt on ISRO EOS-4/RISAT scenes (or realistic synthetic Indian-coast patches) rather than presenting raw benchmark accuracy as if it transfers directly
Isolation-Forest+SVM dual-layer papers	Prove the feasibility of fusing unsupervised AIS anomaly detection with supervised image classification	Attribution often stops at “nearby vessels flagged,” not a ranked, weighted, explainable score	Push attribution further into an explicit, published scoring function (\$6.8), not just a binary anomaly flag

The real structural gap, stated honestly: even in the most advanced existing tools, detection (SAR) and attribution (AIS) remain loosely coupled stages — Cerulean’s own “Source Profiles” ranking feature is a relatively recent addition, and the academic literature (Mouche et al.; the bidirectional-drift papers) still frames *robust two-way source attribution* as an open research question, not solved engineering. That is what

NTRO is actually funding a solution for — not “no one has thought of satellite+AIS.”

A second, India-specific gap: the vessel population most likely to cause chronic small-volume illegal discharge (fishing craft, small coastal traders) is exactly the population least likely to carry continuously-broadcasting AIS. No purely-AIS-based attribution model, however good, can close that gap on its own — it needs to be designed around (e.g., via NavIC-linked small-vessel tracking, or explicit uncertainty flags when AIS coverage is known to be thin), not ignored.

4. Find the Root Cause (RCA)

4.1 The 5 Whys

1. **Why** is it hard to identify who caused a given oil spill? → By the time a slick is detected, the responsible vessel may already be far away, and many other vessels have transited the area.
2. **Why** can't we just check which vessel was nearest at detection time? → Because “at the time” is itself uncertain — the slick was found sometime *after* discharge, and oil keeps drifting/spreading/weathering continuously, so the true origin point *and* origin time have to be estimated, with growing uncertainty the larger that time gap is.
3. **Why** does that uncertainty exist and grow? → Because the physics linking “oil position at detection” back to “oil position at discharge” (advection by wind+current+waves, three-regime spreading, evaporation/emulsification) is only approximately known, forcing/hindcast wind and current data carries its own error, and detection itself only happens at discrete satellite revisit times, not continuously.
4. **Why** isn't continuous monitoring + tight AIS correlation already solved for India? → Because (a) SAR revisit over any one ocean patch is at best every several hours to a couple of days, not continuous; (b) AIS coverage is voluntary/patchy for exactly the small, non-SOLAS vessels most likely to discharge illegally; and (c) India doesn't yet have an EMSA-CleanSeaNet-equivalent standing, automated pipeline fusing SAR + AIS + met-ocean data end-to-end for its own waters.
5. **Why** doesn't that fused pipeline exist yet? → Because it requires integrating data owned by different agencies (ISRO satellites, DG Shipping/ICG AIS networks, IMD/INCOIS ocean-atmosphere forcing) into one automated workflow with a legally-usable output — a systems-integration and institutional problem as much as an algorithms one. This is precisely the kind of gap a hackathon prototype from NTRO is trying to seed a solution for.

Root cause, stated cleanly: the absence of a single automated pipeline that (a) con-

tinuously screens available SAR imagery for oil-like dark spots, (b) discriminates true spills from look-alikes with a documented confidence, (c) reconstructs a physically-grounded drift/weathering history back to a plausible origin window, and (d) cross-references that window against AIS (and ideally non-AIS coastal) vessel traffic with an explainable multi-factor score. Everything else (slow response, weak deterrence, unresolved legal cases) is *downstream impact* of that one root cause.

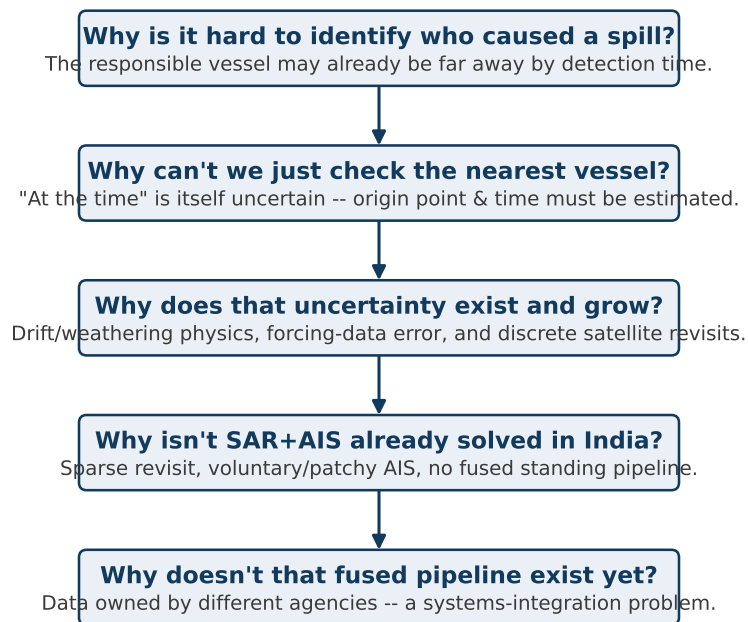


Figure 2: The 5 Whys, chained from symptom to root cause.

4.2 Fishbone (Ishikawa) categorization

- **Technology** — SAR dark-spot/look-alike ambiguity; satellite revisit-time gaps; no unified Indian SAR+AIS+met-ocean pipeline; models trained on non-Indian data may not generalize.
- **Data** — AIS voluntary/spoofable for small non-SOLAS vessels; fragmented ownership across ISRO/DG Shipping/IMD/INCOIS; limited public access to real Indian AIS (the PS's own dataset note points to marinecadastre.gov, which is US-only, and explicitly allows synthetic data as a fallback).
- **Process** — manual analyst-driven workflow doesn't scale to continuous, all-weather, day/night monitoring of ~7,500 km of coastline + ~2.4 million km² of EEZ.
- **People / Institutional** — detection, attribution, and enforcement responsibilities sit across multiple agencies without one owning the fused analytic pipeline end to end.

- **Policy / Legal** — attribution output needs to be defensible enough for enforcement/legal action (MARPOL Annex I violations, NGT proceedings), which raises the bar from “a plausible ML guess” to “an auditable, explainable, evidenced score.”
- **Environment / Physical** — monsoon wind/wave regimes, biogenic slicks from coastal upwelling/aquaculture, river-plume fronts, and dense small-craft fishing traffic all raise Indian-specific look-alike and clutter rates relative to the calmer, better-studied European conditions most published methods were validated on.

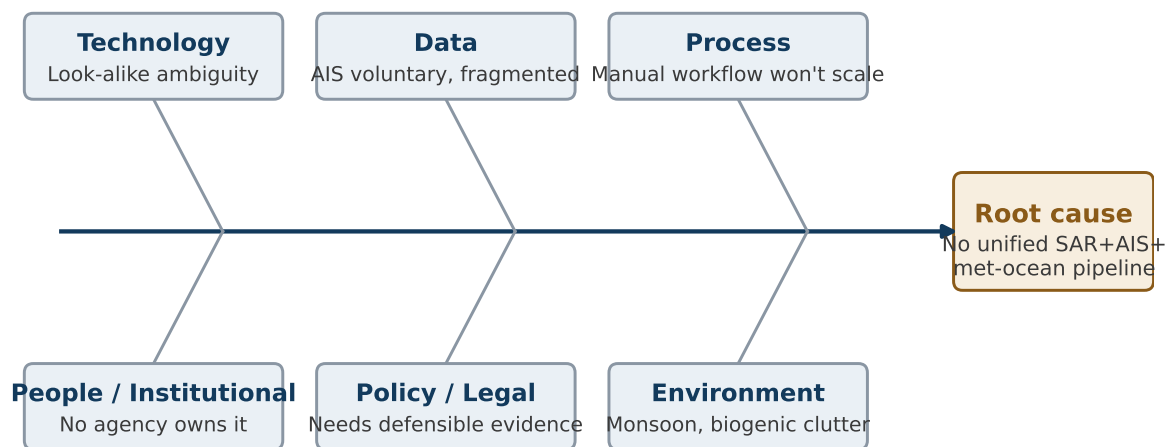


Figure 3: Fishbone (Ishikawa) diagram grouping root causes by category.

4.3 Symptom vs. root cause vs. impact vs. evidence

Symptom

Illegal/accidental oil discharges at sea are frequently never traced to a responsible vessel; response is slow and reactive

Root cause

No unified, automated, India-tuned pipeline linking SAR dark-spot detection → physically-grounded drift/weathering hindcast → AIS (and non-AIS) vessel cross-referencing with an explainable confidence score

Impact	Marine-ecosystem and fisheries damage; weak deterrence against illegal discharge; slow disaster response; legal cases fail for lack of defensible evidence
Evidence	EMSA's decade-plus CleanSeaNet data shows detected-spill rate per monitored area falling as satellite-based enforcement matured; the SAR-oil literature repeatedly flags look-alike discrimination as still-imperfect; backtracking-uncertainty studies (Janeiro et al.) quantify how attribution confidence decays with time-since-discharge — confirming the “many transited vessels, low confidence” symptom has a measurable physical basis, not just an anecdotal one

5. Score the PS

Criterion	Score /10	Reasoning
Impact	8	Real environmental, economic (fisheries/tourism), and legal-enforcement stakes; India's long coastline and heavy tanker traffic make this a live, recurring problem — see the actual May 2025 Kerala MSC ELSA 3 spill, not a hypothetical

Criterion	Score /10	Reasoning
Relevance	9	Matches NTRO's strategic-surveillance mandate directly, and India's existing but fragmented capability (ISRO SAR assets, NavIC vessel tracking, ICG's statutory NOS-DCP role) — a working prototype is genuinely useful to a real sponsoring org
Innovation	5	The core idea (SAR + AIS oil-spill attribution) is well-trodden globally (CleanSeaNet since 2007, Cerulean since ~2023, 15+ years of academic work); real innovation has to come from India-specific tuning, explainable multi-factor scoring, and the non-AIS small-vessel gap — not the base idea
Feasibility	6	Detection (SAR segmentation) is very feasible with public data + pretrained models in 3 weeks; a fully validated, physically-grounded bidirectional drift+weathering+attribution system is multi-month research territory — must be consciously scoped down (§7)

Criterion	Score /10	Reasoning
Scalability	7	The architecture (satellite ingestion → detection → drift model → AIS join → scoring UI) scales conceptually to full-coastline, continuous operation the way CleanSeaNet does, given satellite tasking/compute/AIS access at production scale
Technology fit	8	A data/ML/geospatial-engineering problem — good fit for a CSE/Data-Science team; leverages known building blocks (CNN/U-Net segmentation, Lagrangian particle tracking, anomaly-detection ML) rather than exotic new hardware
Data availability	5	Detection side: good public data (Krestenitis/SOS datasets, free Sentinel-1 via Copernicus, EOS-4 on request from ISRO). Attribution side: real Indian AIS isn't freely public — the PS itself concedes this and permits synthetic AIS, a real constraint to design around

Criterion	Score /10	Reasoning
Prototype feasibility (3 weeks)	6	A convincing demo (pretrained/fine-tuned segmentation + a rule-based/lightweight-ML scorer + a clean map UI) is realistic; a fully validated production-grade bidirectional model is not, and shouldn't be over-promised in the pitch

Overall verdict: a genuinely worthwhile, high-relevance PS with real institutional backing — but not a “quick SVM demo” PS. The innovation ceiling is moderate *unless* you consciously target the specific gaps in §3 (India-tuned look-alike handling, explainable multi-factor attribution, and an honest design response to the AIS-coverage gap) instead of re-deriving what CleanSeaNet/Cerulean already do.

6. Deep Technical & Mathematical Core

This is the “everything, basic to advanced, with the math” section.

6.1 Why oil shows up as a dark patch — SAR ocean-backscatter physics

SAR illuminates the ocean at an angle and measures backscattered energy, called the **Normalized Radar Cross Section (NRCS or σ_0)**, usually reported in decibels (dB).

Over open, wind-roughened, oil-free sea, at moderate incidence angles (roughly 20°–50°), the dominant backscatter mechanism is **resonant Bragg scattering** off short gravity-capillary waves (“Bragg waves”) on the sea surface whose wavelength matches the radar’s, per the resonance condition:

$$\lambda_{\text{Bragg}} = \lambda_{\text{radar}} / (2 \sin \theta)$$

where θ is the local incidence angle and λ_{radar} is the radar wavelength (≈ 5.6 cm for C-band sensors like Sentinel-1, RISAT, or EOS-4).

An oil film — or, crucially, *any* surface-active film, including many non-oil “look-alikes” — increases the local surface’s resistance to stretching (a phenomenon called

Marangoni damping), which suppresses exactly these short Bragg waves. Less roughness at the Bragg wavelength $\rightarrow$ less resonant backscatter $\rightarrow$ the area reads dark against the surrounding rough, oil-free sea.

This is quantified as a **damping ratio**:

$$D = \sigma_{\theta_clean} / \sigma_{\theta_slick}$$

Measurements show D increases with Bragg wavenumber and depends strongly on wind speed — damping is strongest (easiest to detect) at low-to-moderate winds and weaker (harder to detect) at high winds.

Practical detectability window: oil slicks are best seen in SAR at wind speeds of roughly **1.5-7 m/s**.

- Below ~ 1.5 -3 m/s: the *whole* sea surface is calm and naturally dark (low roughness everywhere), so real oil is indistinguishable from ordinary calm water — a major false-positive source (calm patches misread as spills) and false-negative source (real spills invisible against an already-dark background).
- Above ~ 10 -12 m/s: wave action can break up and submerge slicks, and the damping-ratio contrast with the background collapses — real spills become hard to see.

Look-alikes a robust classifier must reject: biogenic surface films (algal exudates, plankton blooms), low-wind/wind-shadow zones (behind land or under rain cells), current-shear fronts, internal waves, rain cells, and — specifically relevant to India — river-plume fronts near major deltas and dense small-craft/aquaculture-pen wake patterns.

Quantitative form — the two-scale Bragg model. The classical Valenzuela (1978) two-scale composite-surface theory writes the NRCS for polarization pp (VV or HH) as proportional to the ocean-wave spectral density *at* the Bragg wavenumber:

$$\sigma_{\theta_pp} \propto \cos^4 \theta \cdot |g_{pp}(\theta, \epsilon_r)|^2 \cdot W(k_{Bragg})$$

where θ is incidence angle, g_{pp} is a polarization- and dielectric-constant-dependent scattering coefficient (ϵ_r is seawater's complex dielectric constant), and $W(k_{Bragg})$ is the value of the ocean-surface wave-height spectrum at the Bragg wavenumber $k_{Bragg} = 2k_{radar} \sin \theta$ ($k_{radar} = 2\pi/\lambda_{radar}$). A surface film changes σ_{θ_pp} almost entirely by suppressing $W(k_{Bragg})$ — i.e., by physically flattening exactly the short waves the radar is resonant with — which is why the damping ratio D in the equation above is really a proxy for the ratio of *damped* to *undamped* wave spectral energy at that one wavenumber, not a generic “brightness” measure. This is also why VV-polarized C-band data (Sentinel-1's default mode, and EOS-4/RISAT's typical modes) is commonly favored for oil-spill work — VV backscatter is more strongly Bragg-dominated at moderate incidence angles than HH, giving cleaner damping con-

trast.

Speckle as a statistical, not just visual, problem. SAR intensity is not deterministic — it follows a multiplicative noise model:

$$I(x) = \sigma_0(x) \cdot n(x)$$

where $n(x)$ is speckle noise. For an L -look intensity image, fully-developed speckle follows a Gamma distribution with shape parameter L and unit mean ($n \sim \text{Gamma}(L, 1/L)$), so $\text{Var}(n) = 1/L$. This is why multi-looking (averaging L independent looks) trades spatial resolution for reduced speckle variance, and why algorithms downstream (thresholding, CFAR, GLCM) are really operating on a *noisy random field*, not a clean image. A standard sanity-check statistic on any homogeneous test patch is the **Equivalent Number of Looks**:

$$\text{ENL} = \mu^2 / \sigma^2$$

(mean-squared over variance, over a spatially homogeneous region) — a low ENL flags residual speckle that a detector hasn't been adequately filtered against, and a useful number to actually report in a hackathon writeup as evidence your preprocessing worked.

6.2 Preprocessing (Stage 0)

1. **Radiometric calibration** — convert raw digital numbers to calibrated σ_0 in dB.
2. **Speckle filtering** — SAR has multiplicative speckle noise from coherent interference. Standard filters (Lee, Refined Lee, Frost, Gamma-MAP) replace each pixel with a locally weighted average that smooths noise while preserving edges. Conceptually, for a Lee-style filter:

$$\text{output} = \text{local_mean} + W \times (\text{pixel} - \text{local_mean})$$

where W is derived from the local coefficient of variation (local std-dev $\div$ local mean) relative to the expected speckle noise's coefficient of variation. Explicitly, the classical Lee filter sets:

$$W = 1 - C_u^2 / C_i^2 \quad (\text{clipped to the range } [0, 1])$$

$$C_u = 1 / \sqrt{L} \quad (\text{coefficient of variation of pure speckle, } L = \text{number of looks})$$

$$C_i = \sigma_{\text{local}} / \mu_{\text{local}} \quad (\text{coefficient of variation actually observed in the local window})$$

When a window is genuinely homogeneous, $C_i \approx C_u$, so $W \rightarrow 0$ and the pixel is replaced by the local mean (heavy smoothing). When a window straddles a real edge (e.g., a slick boundary), $C_i \gg C_u$, so $W \rightarrow 1$ and the original pixel value is preserved almost unchanged — this is exactly the edge-preserving property that makes Lee-family filters the standard choice ahead of dark-spot detection, since blurring a slick's true boundary would corrupt every downstream geometric feature in §6.4.

3. **Land-sea masking** — exclude land pixels so the detector isn't confused by naturally low-backscatter land features (dry sand, radar shadow).

6.3 Dark-spot detection (Stage 1)

(a) Classical adaptive thresholding / CFAR-style. For a moving window over the image, compute the local mean backscatter μ_w , and set a threshold:

$$T = \mu_w - k \text{ (dB)}$$

flagging pixels below T as dark-spot candidates. A **2D Constant False Alarm Rate (CFAR)** refinement adaptively sets k per window so the false-alarm probability stays statistically constant across varying sea states, rather than one fixed global threshold — important because backscatter statistics change with wind and location across a large scene.

Concretely, CFAR chooses T by inverting a target false-alarm probability against the *local* background statistical model, rather than eyeballing a fixed k . If the local sea-clutter background is modeled as Gaussian with mean μ_w and standard deviation σ_w (a simplification — real sea clutter is often better modeled with a Gamma or K-distribution, which have heavier tails, but the Gaussian case is the clean illustrative starting point):

$$P_{fa} = P(I < T \mid \text{background}) = \Phi((T - \mu_w) / \sigma_w)$$

where Φ is the standard normal CDF. Fixing a desired P_{fa} (e.g., 10^{-3}) and solving for T gives:

$$T = \mu_w + \sigma_w \cdot \Phi^{-1}(P_{fa})$$

so the threshold automatically tightens or loosens window-by-window as μ_w/σ_w change across the scene — this is what “constant false-alarm rate” actually means mathematically: the *detection* threshold moves, but the *false-alarm probability* stays fixed. Follow with morphological clustering (drop clusters below a minimum pixel area — single-pixel noise isn't a real slick).

(b) Deep-learning semantic segmentation. Treat this as 5-class pixel labelling (sea / oil / look-alike / ship / land), exactly as Krestenitis et al. framed it, with architectures like U-Net or DeepLabv3+. Two loss functions matter a lot because oil pixels are a small, imbalanced minority class in most scenes:

- **Dice loss:**

$$\text{DiceLoss} = 1 - (2 \cdot \sum(p_i \cdot g_i) + \epsilon) / (\sum p_i + \sum g_i + \epsilon)$$

where p_i is the predicted probability that pixel i is oil, g_i is the ground-truth label (0/1), and ϵ is a small smoothing constant. This directly optimizes overlap between

predicted and true oil regions and is far more robust to class imbalance than plain pixel-wise cross-entropy.

- **Focal loss:**

$$\text{FocalLoss} = -\alpha(1 - p_t)^\gamma \cdot \log(p_t)$$

where p_t is the predicted probability for the *true* class, γ (commonly 2) down-weights the loss from easy, well-classified pixels (the vast majority of “sea” pixels), forcing training attention onto hard, rare pixels (oil/look-alike); α rebalances class-frequency weighting.

Both losses assume the network’s raw output logits z_c (one per class c) have already been converted to probabilities via **softmax**:

$$p_c = \exp(z_c) / \sum_k \exp(z_k)$$

against which plain multi-class **cross-entropy** is simply:

$$\text{CE} = - \sum_c g_c \cdot \log(p_c)$$

($g_c = 1$ for the true class, 0 otherwise) — this is the baseline loss that Dice/Focal are specifically designed to improve on for imbalanced classes.

A useful generalization worth naming explicitly is the **Tversky loss**, which lets you control the false-positive/false-negative trade-off directly rather than treating them symmetrically:

$$\text{Tversky Index} = \text{TP} / (\text{TP} + \alpha \cdot \text{FP} + \beta \cdot \text{FN}) \quad (\alpha + \beta = 1)$$

$$\text{Tversky Loss} = 1 - \text{Tversky Index}$$

Setting $\alpha = \beta = 0.5$ recovers the Dice loss exactly. For this PS specifically, setting $\beta > \alpha$ (e.g., $\alpha=0.3$, $\beta=0.7$) makes the loss penalize **missed real spills (false negatives) more heavily than false alarms (false positives)** — a defensible, explicit design choice to argue for in your pitch, since a missed spill has a much higher real-world cost than a false alarm that a human analyst quickly dismisses.

A combined Dice + Focal (or Dice + weighted cross-entropy, or Tversky + Focal) loss is what the literature reports performing best here.

IoU (Intersection-over-Union) is both a common auxiliary loss term and the standard evaluation metric:

$$\text{IoU} = |P \cap G| / |P \cup G|$$

where P is the predicted-oil pixel set and G is ground truth. Equivalently, in terms of pixel-level True Positives/False Positives/False Negatives:

$$\text{IoU} = \text{TP} / (\text{TP} + \text{FP} + \text{FN})$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{F1} = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall})$$

Note IoU and F1/Dice measure the same underlying overlap but are *not numerically equal* — for the same prediction, $\text{F1} \geq \text{IoU}$ always, because $\text{F1} = 2 \cdot \text{IoU} / (1 + \text{IoU})$ — a common confusion worth getting right when comparing your numbers against published benchmarks like Krestenitis et al.'s.

6.4 Discriminating oil from look-alikes (Stage 2) — the hardest sub-problem

After dark-spot candidates are found, each blob gets a feature vector and is classified oil vs. look-alike:

- **Geometric features** — area A, perimeter P, and a complexity/spreading index:

$$C = P^2 / (4\pi A)$$

(how far the shape departs from a circle — real oil slicks driven by wind/current tend to be elongated and streaky; many biogenic look-alikes are more compact). Also useful: fractal boundary dimension, number of filaments/fingers.

- **Contextual features** — local wind speed at detection time (needed because detectability itself is wind-dependent, per §6.1), proximity to shipping lanes, coincidence with a SAR-visible bright target (a ship) nearby.
- **Texture features via Gray-Level Co-occurrence Matrix (GLCM)**, computed on SAR intensity within the candidate region — standard Haralick features:

$$\text{Contrast} = \sum_{i,j} (i-j)^2 \cdot P(i,j)$$

$$\text{Homogeneity} = \sum_{i,j} P(i,j) / (1 + |i-j|)$$

$$\text{Energy (ASM)} = \sum_{i,j} P(i,j)^2$$

$$\text{Entropy} = -\sum_{i,j} P(i,j) \cdot \log(P(i,j))$$

$$\text{Dissimilarity} = \sum_{i,j} |i-j| \cdot P(i,j)$$

$$\text{Correlation} = \sum_{i,j} (i-\mu_i)(j-\mu_j) \cdot P(i,j) / (\sigma_i \cdot \sigma_j)$$

where $P(i,j)$ is the normalized co-occurrence probability of intensity pairs (i,j) at a chosen pixel offset, and $\mu_i, \sigma_i (\mu_j, \sigma_j)$ are the mean and standard deviation of the row (column) marginal distributions of P . Real oil slicks tend toward smoother, more homogeneous internal texture (higher homogeneity/energy, lower entropy/contrast/dissimilarity) than many look-alikes — not a perfectly clean separator alone, which is exactly why it's fed into a classifier rather than used as a hard rule.

- **Classifiers** historically used on this feature set: Random Forest, SVM, shallow ANN/MLP — these often outperform end-to-end deep segmentation *for this specific discrimination step*, because the hand-crafted features above encode do-

main knowledge a generic CNN would otherwise need far more data to learn from scratch. Concretely, a kernel-SVM makes its oil-vs-look-alike decision as:

$$f(x) = \text{sign}(\sum_i \alpha_i \cdot y_i \cdot K(x_i, x) + b)$$

where x is a candidate blob's feature vector (geometry + GLCM + context from above), (x_i, y_i) are labeled training blobs, K is a kernel function (an RBF kernel $K(x_i, x) = \exp(-\gamma \|x_i - x\|^2)$ is the usual default for this kind of tabular feature-vector classification), and α_i, b are learned from training data. The appeal for a hackathon: it trains fast on a few hundred labeled blobs (unlike a CNN, which needs thousands of labeled pixels) and its support vectors/weights are directly inspectable — useful for the explainability argument made throughout this dossier.

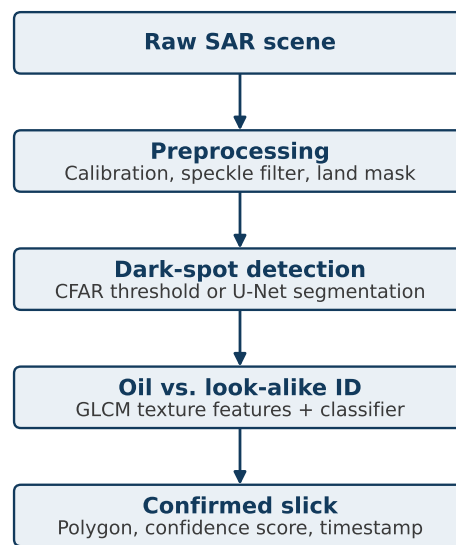


Figure 4: The detection pipeline (Sections 6.1-6.4) end to end.

6.5 Oil spreading & weathering (fate)

Once a candidate spill is confirmed, you need a fate/transport model both to forecast where it's going and to hindcast where it came from.

(a) Spreading — Fay's classical three-regime model. Describes how a slick's own physics (gravity, inertia, viscosity, surface tension) grows its area even with zero current/wind, from the moment oil hits water:

- **Regime I — gravity-inertia** (seconds to minutes): gravity resisted by the oil's own inertia.

$$D_1(t) \propto (\Delta g \cdot V \cdot t^2)^{(1/4)}$$

- **Regime II — gravity-viscous** (minutes to hours): gravity now resisted by viscous drag from the underlying water.

$$D_2(t) \propto (\Delta g \cdot V^2 \cdot t^{1.5} / \sqrt{\nu_w})^{1/6}$$

- **Regime III — surface-tension-viscous** (hours to days, thin sheens): the spreading coefficient σ dominates, resisted by water viscosity.

$$D_3(t) \propto (\sigma^2 \cdot t^3 / (\rho_w^2 \cdot \nu_w))^{1/4}$$

where $\Delta = (\rho_w - \rho_o)/\rho_w$ is the relative density difference between seawater (ρ_w) and oil (ρ_o), g is gravitational acceleration, V is spilled volume, t is elapsed time, ν_w is seawater's kinematic viscosity, and σ is the oil-water-air spreading coefficient.

Important caveat, straight from the literature: Fay's original proportionality constants (commonly cited around 1.14 / 1.45 / 1.6 for the three regimes) vary by source, and the model is well documented to *under-predict* real spreading. Lehr et al. (1984) and later Chebbi (2014) proposed corrections — e.g., letting the slick elongate along the wind direction instead of staying circular. Present these as the standard starting point, not exact truth — that nuance itself is worth a slide in your pitch.

Two extra pieces of math worth having on hand:

- **Film thickness from mass/volume conservation.** Treating the (roughly constant, ignoring weathering losses for the moment) spilled volume V as spread over the model's own predicted area $A(t) = \pi \cdot D(t)^2 / 4$, the mean film thickness at any time is simply:

$$h(t) = V / A(t)$$

This matters directly for detectability (§6.1): once $h(t)$ drops to the sub-micron “sheen” range (roughly $<0.1 \mu\text{m}$), the film becomes too thin to meaningfully damp Bragg waves, and the slick can vanish from SAR well before it has actually dispersed — a genuine detection-model limitation worth stating explicitly rather than assuming “detected” and “gone” are the same thing.

- **Regime transition times.** Fay's three regimes aren't separated by a fixed clock time — they're separated by whichever forces dominate at a given slick size, and the transition time between regime i and regime $i+1$ is found by setting $D_i(t) = D_{i+1}(t)$ and solving for t . For example, equating Regime I and Regime II gives a transition time t_{12} that scales with spill volume and fluid properties (larger spills stay in the inertia-dominated Regime I longer); equating Regime II and III similarly gives t_{23} . In practice this means: don't apply Regime III's thin-sheen physics to a size-classed large tanker spill in its first hour, and don't apply Regime I's early-time physics to a multi-day-old slick — matching the regime to the actual elapsed time (or, better, solving for which regime currently applies from the transition-time equations) is a basic correctness check your drift module should

run before every particle-propagation step.

(b) Advection — where the whole slick physically moves:

$$V_{oil} = V_{current} + \alpha \cdot V_{wind(10m)} + V_{Stokes}$$

where $V_{current}$ is near-surface ocean current (from INCOIS/Copernicus Marine re-analysis), $V_{wind(10m)}$ is wind speed at the standard 10 m reference height, α is the empirical **wind drift factor** (widely cited range 2.5–4% of wind speed, ~3–3.5% commonly used operationally), and V_{Stokes} is the wave-induced Stokes-drift contribution. An empirical deflection angle (a few degrees up to a few tens of degrees, generally to the right of the wind in the Northern Hemisphere due to Coriolis/Ekman-type effects) is sometimes applied — exact values differ across operational models (NOAA/GNOME-style vs. European Seatrack-Web-style), so treat these as tunable parameters, not one hard-coded textbook number.

Horizontal dispersion (turbulent spreading beyond deterministic Fay growth) is usually added as a random-walk/Fickian diffusion term on top of this advection — exactly how GNOME and OpenDrift implement it internally.

(c) Weathering — mass/property change over time, which matters both for how visible a slick stays in SAR and for how far/fast it can travel:

- **Evaporation** — light crude can lose up to ~75% of volume within days; heavy/residual oils lose <10% in the same period. A simple, well-validated empirical form (Fingas, building on Mackay & Matsugu / Stiver & Mackay):

$$\%evaporated = C(T) \cdot \ln(t)$$

with a commonly cited fitted variant:

$$\%evaporated = 0.165 \times (\%D) \times \ln(t)$$

where %D is the percentage (by weight) of the oil that distills at 180°C (a lab-measured, oil-specific constant, available from databases like NOAA's ADIOS Oil Database), t is time in minutes, and $C(T)$ is an oil- and temperature-dependent empirical constant.

The underlying physical process this empirical fit is standing in for is a **mass-transfer-limited evaporative flux**, the Stiver & Mackay (1984) formulation:

$$N_{evap} = K_m \cdot A \cdot (P_{vap} / R \cdot T)$$

where N_{evap} is the molar evaporation rate, A is slick area, P_{vap} is the vapor pressure of the bulk liquid, R is the gas constant, T is absolute temperature, and K_m is a wind-driven mass-transfer coefficient — commonly parameterized (Mackay & Matsugu, 1973) with an empirical wind-speed dependence of roughly $K_m \propto U^{0.78}$ (i.e., evaporation accelerates faster than linearly with wind speed, but sub-quadratically) times a Schmidt-number correction for the diffusing vapor. This is the equation that actually explains *why* the simple $\ln(t)$ fit above works: as the lighter, more volatile

fractions evaporate first, P_{vap} of the remaining bulk liquid keeps dropping, which is exactly what produces the observed logarithmic-in-time (rather than linear) mass loss curve.

- **Emulsification** — oil takes up water to form “mousse,” expanding volume 3–5× and raising viscosity by up to ~3 orders of magnitude; stable emulsions typically contain 60–85% water. This changes drift response (higher windage) and dramatically slows further spreading/evaporation — a key reason an old, weathered slick behaves very differently in a drift model than a fresh one, and why estimating oil “age” from the SAR signature matters for setting realistic weathering initial conditions.

The rate of water uptake is commonly modeled with **Mackay’s empirical water-uptake equation**:

$$dY/dt = K_A \cdot (1 + U)^2 \cdot (1 - Y / Y_{\text{max}})$$

where Y is the water-volume fraction currently in the emulsion, U is wind speed, K_A is a small empirical rate constant (oil-specific, typically of order 10^{-6}), and Y_{max} (≈ 0.7 – 0.85 depending on the oil’s asphaltene/resin content) is the maximum stable water fraction that oil type can hold — note the $(1 - Y/Y_{\text{max}})$ term means uptake naturally slows and saturates as the emulsion approaches its stability limit, rather than growing without bound.

The resulting viscosity jump is often approximated with the **Mooney equation** for the viscosity of a suspension/emulsion as a function of dispersed-phase (water) fraction:

$$\mu_{\text{emulsion}} = \mu_{\text{oil}} \cdot \exp(2.5 \cdot Y / (1 - k \cdot Y))$$

where μ_{oil} is the fresh (pre-emulsified) oil’s viscosity and k (a “self-crowding” factor, typically ≈ 1.2 – 1.5) captures how droplets interfere with each other’s flow as Y grows. This is the equation that quantitatively explains the “up to ~3 orders of magnitude” viscosity increase mentioned above, and it’s directly consequential for drift modeling: a higher-viscosity mousse resists the wind-drift and Fickian-diffusion terms in §6.6 differently than fresh oil, so a hindcast that ignores emulsification will systematically misplace an older slick’s estimated origin.

- **Dissolution and photo-/microbial oxidation** are typically minor mass-balance terms on the surface-slick timescale relevant here (hours–days) but matter for longer-term ecological modeling; most fast operational pipelines simplify these into a slow bulk-loss term.

6.6 Backward hindcasting & forward tracking — the Lagrangian-particle math

The engine underneath GNOME and OpenDrift (and what a from-scratch prototype needs at a basic level) is **Lagrangian particle tracking**: represent the slick as many discrete particles, and integrate each one’s position over time according to:

$$dx/dt = u(x, t)$$

where $u(x, t)$ is the total local velocity field (V_{oil} from §6.5b, built from gridded current + wind forcing interpolated in space and time), integrated numerically — simple forward-Euler ($x(t+\Delta t) = x(t) + u(x, t) \cdot \Delta t$) for a first pass, or, for real stability with the larger timesteps a hackathon’s compute budget needs, a **4th-order Runge-Kutta (RK4)** step:

$$\begin{aligned} k1 &= u(x_n, t_n) \\ k2 &= u(x_n + (\Delta t/2) \cdot k1, t_n + \Delta t/2) \\ k3 &= u(x_n + (\Delta t/2) \cdot k2, t_n + \Delta t/2) \\ k4 &= u(x_n + \Delta t \cdot k3, t_n + \Delta t) \\ x_{n+1} &= x_n + (\Delta t/6) \cdot (k1 + 2k2 + 2k3 + k4) \end{aligned}$$

This is the same numerical scheme GNOME/OpenDrift use internally, and it’s noticeably more accurate than forward-Euler for the same computational budget when the velocity field curves quickly (e.g., near coastlines or eddies) — worth citing explicitly if a judge asks “why not just Euler-integrate.”

The whole problem is equivalently expressible in **Eulerian (field) form** rather than as individual particle trajectories — useful context because it’s the form most oil-spill-model documentation (and the underlying oceanographic literature) actually states the governing physics in:

$$\partial C / \partial t + \nabla \cdot (u \cdot C) = \nabla \cdot (K \cdot \nabla C) + S$$

where $C(x, t)$ is the oil concentration field, u is the advective velocity field (§6.5b), K is a turbulent-diffusivity tensor, and S is a source/sink term (weathering losses from §6.5c). The Lagrangian particle-tracking approach above is a Monte-Carlo way of solving exactly this PDE without discretizing the whole ocean into a grid — each particle is a “sample” of the concentration field, and their statistical spread over time reproduces the $\nabla \cdot (K \nabla C)$ diffusion term via a stochastic random-walk step added on top of the deterministic advection:

$$x(t+\Delta t) = x(t) + u(x, t) \cdot \Delta t + \sqrt{(2 \cdot K \cdot \Delta t)} \cdot \xi, \quad \xi \sim N(0, 1)$$

This is precisely how GNOME and OpenDrift implement “horizontal dispersion” in practice (mentioned qualitatively above) — now with the actual stochastic differential equation behind it. The $\sqrt{(\Delta t)}$ (not Δt) scaling of the noise term is the signature of Fickian/Brownian-type diffusion and is worth getting right if you implement this

from scratch, since using a Δt -scaled random term instead would silently make your particle cloud spread at the wrong rate.

- **Backward hindcasting** — to estimate where/when a detected slick likely originated, integrate the same equation with time running backward, seeding many particles across the observed slick's extent. The resulting cloud of backward-tracked positions at each candidate earlier time defines a probable **origin footprint** (a spread, not a single point — uncertainty naturally grows the cloud), which you compare against AIS vessel positions at that past time.
- **Forward tracking (Mouche-et-al.-style)** — instead of backtracking one slick, forward-simulate a hypothetical continuous discharge from every candidate vessel's *actual* AIS-recorded track, and check whose simulated footprint best overlaps the observed slick. This avoids compounding all uncertainty into a single backward run, at the cost of needing to simulate many vessels instead of one slick.
- **Bidirectional fusion** — run both, and only treat a vessel as a strong candidate when its forward-simulated footprint *and* the slick's own backward-hindcast footprint agree. This directly targets the many-to-one/one-to-many ambiguity flagged as the real open gap in §3.
- **Uncertainty quantification** — because forcing data has error and discharge time is unknown, a single deterministic backtrack isn't defensible as evidence. An ensemble/Monte-Carlo approach perturbs the input forcing (and/or assumed discharge time) across many runs and reports a probability-density **search cone**, not one line — standard practice in real tools (GNOME's uncertainty mode, Seatrack Web) and should be treated as a required feature, not optional polish, if this PS is meant to produce evidence-grade output.

6.7 AIS data engineering

- **Message basics** — Class A vessels (SOLAS-mandated, generally >300 GT international) broadcast dynamic position reports (Message Types 1/2/3): MMSI (unique ID), lat/lon, Speed Over Ground (SOG), Course Over Ground (COG), true heading, rate-of-turn, navigational status, UTC timestamp — roughly every 2–10 s depending on speed/manoeuvring. Static/voyage data (Message Type 5: IMO number, call sign, name, ship type, dimensions, draught, destination, ETA) broadcasts every few minutes. Class B (smaller/non-SOLAS, optional carriage) uses Message Types 18/19 at lower report rates.
- **Track reconstruction** — raw AIS points are irregularly spaced and noisy. Standard practice: interpolate (linear or cubic-spline) between reports, or better, apply a **Kalman filter**, which recursively combines a motion-prediction step with each new noisy observation, naturally handling irregular reporting intervals and producing both a smoothed track *and* an uncertainty estimate at any

queried time — directly useful for estimating where a vessel probably was at your hindcast’s candidate discharge time even if it didn’t broadcast exactly then. Concretely, with state vector $x = [\text{lat}, \text{lon}, v_{\text{lat}}, v_{\text{lon}}]^T$ (position + velocity), the filter alternates:

Prediction:

$$\begin{aligned}\hat{x}_k |_{k-1} &= F \cdot \hat{x}_{k-1} |_{k-1} \\ P_k |_{k-1} &= F \cdot P_{k-1} |_{k-1} \cdot F^T + Q\end{aligned}$$

Update (on receiving a new AIS fix z_k):

$$\begin{aligned}K_k &= P_k |_{k-1} \cdot H^T \cdot (H \cdot P_k |_{k-1} \cdot H^T + R)^{-1} \\ \hat{x}_k |_k &= \hat{x}_k |_{k-1} + K_k \cdot (z_k - H \cdot \hat{x}_k |_{k-1}) \\ P_k |_k &= (I - K_k \cdot H) \cdot P_k |_{k-1}\end{aligned}$$

where F is the constant-velocity motion-model matrix (propagates position by velocity $\times \Delta t$), H maps state to the observed quantities (AIS gives position directly, so H just selects the lat/lon components), Q is process noise (how much a real vessel’s motion can deviate from constant-velocity — manoeuvring, currents, etc.), R is AIS measurement noise, and K_k — the **Kalman gain** — automatically weights the new observation against the prediction in proportion to their relative uncertainties. The output $P_k |_k$ at any queried time is exactly the “uncertainty estimate” mentioned above, and is what should feed the CPA/TCPA calculation in §6.8 rather than treating an interpolated AIS position as if it were exact.

- **AIS-gap (“dark”) analysis** — flag any interval where a previously-reporting vessel’s AIS feed stops for an unusually long duration relative to its normal cadence. A well-documented behavioral red flag (used by Windward, Skylight/Cerulean’s “AIS-off events”), since intentionally disabling AIS during illegal discharge is a known evasion pattern.
- **Spatiotemporal search** — restrict AIS comparison to a **space-time cone**: the union of all positions reachable from the slick’s estimated origin footprint (§6.6) within a physically plausible window before detection. This is both a computational necessity and a principled way to bound false positives.

6.8 Vessel-source attribution scoring — the real novel-contribution surface

This is where a strong team actually differentiates (per §3’s gap analysis). A defensible design combines several independently computed sub-scores into one explainable ranking, rather than a single opaque ML output:

- **Distance/overlap metrics:**
 - **Haversine great-circle distance:**

$$d = 2r \cdot \arcsin(\sqrt{(\sin^2(\Delta\phi/2) + \cos \phi_1 \cdot \cos \phi_2 \cdot \sin^2(\Delta\lambda/2))})$$

where ϕ is latitude, λ is longitude, r is Earth's radius ($\sim 6,371$ km) — the basic building block for any proximity feature.

- **Closest Point of Approach (CPA) / Time to CPA (TCPA):** treating a candidate vessel's track and the slick's drifted position as two moving points with relative position vector r and relative velocity vector v :

$$\text{TCPA} = -(r \cdot v) / |v|^2$$

$$\text{CPA} = |r + v \cdot \text{TCPA}|$$

Standard collision-avoidance math, repurposed to score how close a vessel's actual track came to the slick's backward-hindcast origin footprint at the relevant time.

- **Drift-overlap score** — the fractional spatial overlap (an IoU-style ratio) between a candidate vessel's forward-simulated discharge footprint (§6.6) and the actually observed slick polygon.
- **Behavioral-anomaly features** (feeding an unsupervised model like **Isolation Forest**, or a density-based one like **DBSCAN**, per §2.4/§2.6): sudden unexplained speed drops, prolonged loitering, erratic/zig-zag course changes, AIS-gap duration and timing (§6.7), deviation from the vessel's own historical route baseline for that leg.

Isolation Forest, concretely: it builds many random binary trees that recursively split the feature space, and anomalies — being rare and “different” — tend to get isolated (end up alone in their own leaf) in far fewer splits than normal points. The anomaly score for a point x is:

$$s(x, n) = 2^{-(E[h(x)] / c(n))}$$

where $h(x)$ is the path length (number of splits) needed to isolate x in a given tree, $E[h(x)]$ is that path length averaged across all trees in the forest, and $c(n)$ is the average path length of an *unsuccessful* search in a Binary Search Tree of n points — a normalization constant, $c(n) = 2 \cdot H(n-1) - 2(n-1)/n$, with $H(i)$ the harmonic number ($H(i) \approx \ln(i) + 0.5772$, using the Euler-Mascheroni constant). Scores near 1 flag strong anomalies (isolated in very few splits); scores near 0.5 or below indicate normal points — a clean, unsupervised way to flag a vessel's zig-zagging or loitering leg without needing any labeled “this was suspicious” training data.

DBSCAN, if used instead/additionally for spatial clustering of AIS positions (e.g., to find dense “normal shipping lane” clusters vs. sparse outlier tracks): a point p is a **core point** if its ϵ -neighborhood contains at least MinPts other points,

$$p \text{ is a core point } \Leftrightarrow |N_{\epsilon}(p)| \geq \text{MinPts}, \quad \text{where } N_{\epsilon}(p) = \{ q : \text{dist}(p, q) \leq \epsilon \}$$

points reachable from a core point form a cluster; points that are neither core nor reachable are labeled noise/outliers — directly usable to flag a vessel whose track sits outside all normal-traffic clusters as behaviorally anomalous.

- **A weighted multi-criteria fusion score:**

$$\text{Score}(\text{vessel}_i) = w_1 \cdot f(\text{CPA}_i) + w_2 \cdot f(\text{TCPA}_i) + w_3 \cdot f(\text{drift_overlap}_i) \\ + w_4 \cdot f(\text{AIS_gap_anomaly}_i) + w_5 \cdot f(\text{behavioral_anomaly}_i)$$

where each $f(\cdot)$ is a normalized (0–1) sub-score. A concrete, easy-to-defend choice is a **Gaussian-kernel transform** on each raw distance/time value:

$$f(d) = \exp(-d^2 / 2\sigma^2)$$

where d is the raw CPA distance (or TCPA time-gap, or drift-overlap deficit) and σ is a tunable “how forgiving” bandwidth — small σ sharply penalizes anything beyond a tight radius, large σ gives a gentler falloff; picking σ from the drift model’s own uncertainty cone (§6.6) rather than an arbitrary number is the principled move, since it ties the scoring function’s tolerance directly to how much physical uncertainty the hindcast actually has. Weights w are either hand-tuned with documented domain justification (simplest, most explainable — good for both a hackathon demo and legal defensibility) or learned via logistic regression/gradient boosting if enough labeled historical cases exist.

A genuinely **Bayesian** version would instead compute, for each candidate:

$$P(\text{vessel}_i \text{ is source} \mid \text{evidence}) \propto P(\text{evidence} \mid \text{vessel}_i \text{ is source}) \cdot P(\text{vessel}_i \text{ is source})$$

normalized across all candidates. Worked out concretely, a plausible (simplified) likelihood term for just the spatial-proximity evidence might itself take the same Gaussian form as above,

$$P(\text{CPA}_i \mid \text{vessel}_i \text{ is source}) = \exp(-\text{CPA}_i^2 / 2\sigma^2)$$

with independent evidence channels (CPA, drift-overlap, behavioral anomaly) multiplied together under a naive-independence assumption to form the full likelihood, and the prior $P(\text{vessel}_i \text{ is source})$ set uniformly across all candidates in the space-time cone (or non-uniformly, e.g., down-weighted for vessel types/cargo unlikely to carry the detected oil’s apparent grade). This is more rigorous, but needs careful likelihood modeling — and an honest independence assumption — that a hackathon timeline usually can’t fully validate. A well-justified weighted-sum is a reasonable, *honest* scope for the prototype, with full Bayesian fusion explicitly named as future work rather than glossed over.

- **Present the final output as a ranked shortlist showing each vessel’s sub-scores**, not just one final number — this transparency is exactly what differentiates the tool from a black box and directly answers the explainability gap identified in §3.

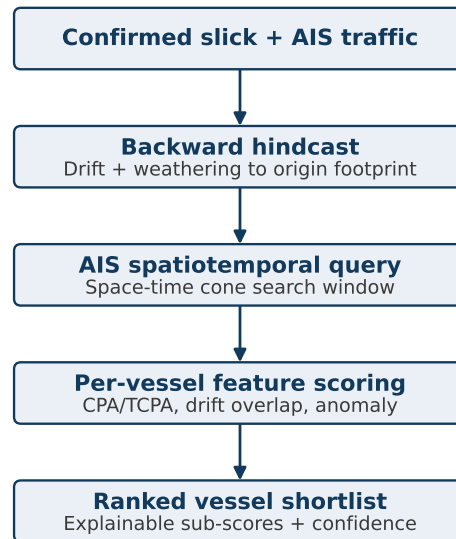


Figure 5: The attribution pipeline (Sections 6.5-6.8) end to end.

6.9 End-to-end system architecture (conceptual)

1. **Ingestion** — scheduled pull of new SAR scenes (Sentinel-1 via Copernicus Open Access Hub, or EOS-4/RISAT via ISRO on request) over the area of interest, plus AIS stream (real if available, else synthetic per the PS’s own guidance) and met-ocean forcing (wind/current reanalysis from INCOIS/Copernicus Marine Service).
2. **Detection service** — preprocessing (§6.2) → dark-spot detection (§6.3) → oil-vs-look-alike discrimination (§6.4) → confirmed slick polygon + confidence + timestamp.
3. **Fate/transport service** — seed particles on the confirmed polygon → backward hindcast (§6.6) with ensemble uncertainty → candidate origin footprint(s) across a window of candidate discharge times.
4. **Attribution service** — spatiotemporal AIS query (§6.7) over the candidate footprint/time window → per-vessel CPA/TCPA/drift-overlap/behavioral features (§6.8) → fused ranked score.
5. **Visual interface** (explicitly requested by the PS) — map view showing the slick, its backward-hindcast uncertainty cone, all candidate vessel tracks color-coded by score, and a ranked table with sub-score breakdown and an exportable evidence report. That last feature — an exportable, structured evidence report — is small to build but directly serves the enforcement/legal stakeholder identified in §1.2.

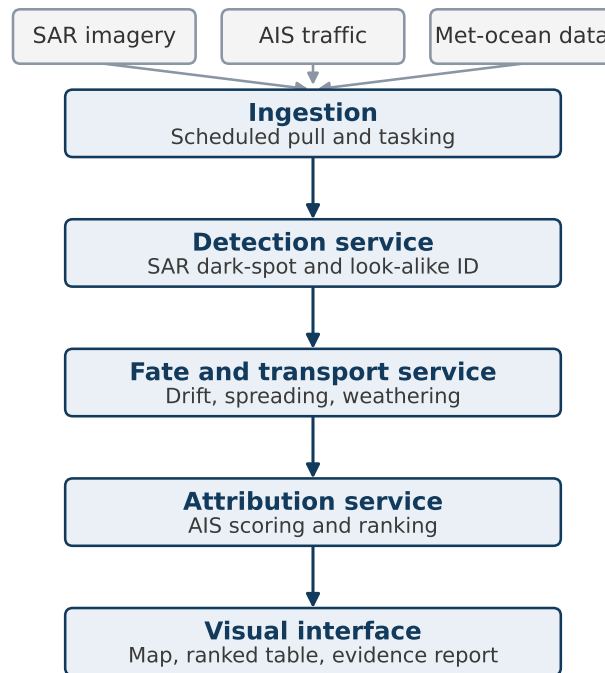


Figure 6: End-to-end system architecture – all five services in sequence.

6.10 Evaluation metrics to actually report

- **Detection/segmentation:** IoU, Dice/F1, pixel accuracy, and specifically per-class recall for the “oil” class alone — overall accuracy is misleading given class imbalance. Report “oil” and “look-alike” performance separately, since confusing them in opposite directions has very different real-world consequences (a missed real spill vs. a false accusation).
- **Attribution:** ground truth (“which vessel actually did it”) is rarely available for real spills, so use Top-1/Top-k accuracy and **Mean Reciprocal Rank:**

$$\text{MRR} = (1/|Q|) \cdot \sum_q (1 / \text{rank_of_correct_vessel_in_query_q})$$

across a labeled test set Q you construct (synthetic injected “true” polluter cases, or documented real incidents published by EMSA/Cerulean), plus a calibration check — does a vessel scored “80% confidence” actually turn out correct about 80% of the time across many synthetic trials? Calibration matters enormously for a tool meant to inform real enforcement decisions. If reporting detection performance scene-by-scene rather than per-pixel, **mean Average Precision** is the standard summary:

$$\text{mAP} = (1/N) \cdot \sum_i \text{AP}_i$$

where AP_i is the area under the precision-recall curve for scene/class i and N is the

number of scenes/classes averaged over.

6.11 Dimensionless numbers behind the spreading regimes

Fay’s three regimes (§6.5a) aren’t an arbitrary taxonomy — they correspond to which term dominates a force balance, which is exactly what fluid-dynamics **dimensionless numbers** are built to express. Worth having these on hand, since they generalize the “which regime am I in” question beyond just plugging into Fay’s formulas:

Reynolds number: $Re = \rho \cdot U \cdot L / \mu$ (inertial forces ÷ viscous forces)
 Froude number: $Fr = U / \sqrt{g \cdot L}$ (inertial forces ÷ gravitational forces)
 Weber number: $We = \rho \cdot U^2 \cdot L / \sigma$ (inertial forces ÷ surface-tension forces)

where ρ is fluid density, U a characteristic velocity (e.g., the slick’s own spreading rate dD/dt), L a characteristic length (slick diameter D), μ dynamic viscosity, g gravity, and σ surface tension. Regime I (gravity-inertia) is the regime where Fr terms matter most and Re is high (inertia beats viscosity); Regime II (gravity-viscous) is where Re drops enough that viscous drag starts to matter; Regime III (surface-tension-viscous) is where We becomes small enough that surface tension, not gravity, drives further spreading. Framing the regime transitions this way — rather than only via the transition-time algebra in §6.5a — is the more “textbook fluid dynamics” way to justify your model’s regime-switching logic if a judge with a mechanical/aerospace background pushes on it.

6.12 Wind-drift deflection and the Coriolis connection

§6.5b mentioned that the wind-drift term in the advection equation is sometimes deflected by an empirical angle “to the right of the wind in the Northern Hemisphere.” The physical reason is the **Coriolis effect**, quantified by the Coriolis parameter:

$$f = 2 \cdot \Omega \cdot \sin(\phi)$$

where $\Omega = 7.292 \times 10^{-5}$ rad/s is Earth’s rotation rate and ϕ is latitude. In classical Ekman-layer theory, a steady wind stress on the ocean surface drives a current that spirals with depth, with the *surface* current deflected roughly 45° from the wind direction (to the right in the Northern Hemisphere, left in the Southern) under idealized deep-water, steady-state assumptions — the smaller deflection angles (a few degrees to a couple of tens of degrees) actually used in most *operational* oil-drift models are empirically tuned reductions of this idealized 45°, since real, shallow, coastal, non-steady conditions (exactly what much of the Indian coastline presents) don’t fully develop a classical Ekman spiral. The direct practical payoff: because f depends on $\sin(\phi)$, the “correct” deflection angle is not a universal constant — it should, in principle, vary somewhat with the latitude of the spill (near the equator, $f \rightarrow 0$ and the

deflection effect weakens substantially), which is one more reason to treat it as a tunable, region-calibrated parameter rather than a hard-coded textbook number, exactly as flagged in §6.5b.

7. Suggested Prototype Scope for a 3-Week Sprint

Given your team’s compressed 3-week timeline, be explicit that the full pipeline above is multi-month research territory, and scope the demo honestly:

Tier 1 — must-have, achievable:

- Fine-tune/run a segmentation model on the public Krestenitis or SOS dataset (or ESA SNAP’s built-in adaptive-threshold operator as a faster fallback) to detect + classify oil vs. look-alike on sample Sentinel-1 scenes.
- Build a synthetic-AIS generator (since real Indian AIS isn’t public) validated to look statistically like real regional traffic.
- Implement a basic forward/backward advection model (wind-drift-factor + current, without full Fay-regime detail) to connect slick ↔ candidate vessels.
- Implement the weighted multi-criteria scoring (§6.8) with CPA/TCPA + drift overlap.
- Build the map + ranked-table visual interface with an exportable evidence report.

Tier 2 — stretch goals, if time remains:

- Add Fay-regime spreading and Mackay/Fingas-style weathering for a more physically faithful hindcast.
- Add Isolation-Forest-based AIS behavioral-anomaly scoring.
- Add Monte-Carlo uncertainty cones instead of a single deterministic backtrack.

Tier 3 — explicitly future work, say so in the pitch rather than overclaiming:

- Full Bayesian fusion, real-time production-scale ingestion, legal-grade chain-of-custody certification, validated cross-sensor (ALOS/RISAT/EOS-4) generalization.

TIER 1 -- must-have, achievable

- Fine-tune / run segmentation on Krestenitis or SOS dataset (or ESA SNAP fallback)
- Synthetic-AIS generator validated against realistic regional traffic
- Basic forward/backward advection model (wind-drift factor + current)
- Weighted multi-criteria scoring: CPA/TCPA + drift overlap
- Map + ranked-table UI with exportable evidence report

TIER 2 -- stretch goals, if time remains

- Fay-regime spreading + Mackay/Fingas-style weathering
- Isolation-Forest-based AIS behavioral-anomaly scoring
- Monte-Carlo uncertainty cones, not one deterministic backtrack

TIER 3 -- explicitly future work (say so in the pitch)

- Full Bayesian fusion
- Real-time production-scale ingestion
- Legal-grade chain-of-custody certification
- Validated cross-sensor (ALOS/RISAT/EOS-4) generalization

Figure 7: The three-tier scope split for a 3-week sprint.

Being explicit about this tiering in the actual pitch is itself a strength — judges tend to penalize teams that quietly overclaim a research-grade capability as “done” in a hackathon demo, and reward teams that clearly understand where the real research frontier is, which is exactly what §§2–6 above establish.

8. Key Sources for Further Reading

- Krestenitis et al. (2019), *Oil Spill Identification from Satellite Images Using Deep Neural Networks* — Remote Sensing 11(15):1762 (DOI: 10.3390/rs11151762)
- EMSA CleanSeaNet — <https://www.emsa.europa.eu/csn-menu.html>
- NOAA GNOME/PyGNOME/ADIOS — <https://response.restoration.noaa.gov/oil-and-chemical-spills/oil-spills/response-tools/gnome-suite-oil-spill-modeling.html>
- OpenDrift / OpenOil (MET Norway) — <https://opendrift.github.io/>
- SkyTruth Cerulean — <https://skytruth.org/cerulean>
- ESA SNAP Oil Spill Detection operator — step.esa.int – SNAP toolbox docs
- PierSight (Indian startup, INDUS-X Maritime Challenge winner) — <https://piersight.space/blog/wins-us-india-challenges-indus-x-for-oil-spill-detection>

- “Oil in the Sea III: Inputs, Fates, and Effects” (National Academies Press, 2003) — Chapter 4, Behavior and Fate of Oil — <https://www.ncbi.nlm.nih.gov/books/NBK220700/>
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Prepared for Arnav’s SIH 2026 PS 26143 deep-dive. Tiering in §7 is a suggestion, not a constraint — adjust to your team’s actual bandwidth and each member’s strengths.